

# Mathematical modeling of probabilistic processes in railway transport delays on the Almaty-1-Aktogay section

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## 1 Abstract

Forecasting rail transport delays is critical for operational efficiency, particularly on single-track sections such as the Almaty-1–Aktogay route. This 557-km corridor, comprising 40 major and minor stations, serves as a vital artery for freight transit between Central Asia, Siberia, and China. Due to its single-track nature, traffic management relies heavily on passing loops; consequently, any delay triggers a cascading "knock-on" effect across the network. Our field expedition revealed that the Almaty-1–Ushtobe sub-section is characterized by high operational density due to frequent shunting operations at intermediate stations, significantly increasing stochastic delay risks. Furthermore, the transition to semi-automatic blocking systems beyond Ushtobe—signaled by single-lens traffic lights—introduces additional constraints on throughput and human-factor reliability. Adverse weather conditions further exacerbate these challenges. To address these complexities, this study proposes a probabilistic model using Gaussian Processes (GP) to map arrival and departure delays. Experimental results demonstrate that while modern locomotives (360,000 kg mass) largely neutralize the impact of track gradients, GP effectively captures the high-variance uncertainty (noise) generated by station-level bottlenecks and signaling limitations, providing a robust "uncertainty envelope" for logistical risk assessment and cost optimization.

## 2 Introduction

The Almaty-1–Aktogay railway section, spanning 557 km, is a vital segment of the historic 2,375-km Turksib railway line. As a primary artery for the Trans-Kazakhstan corridor, Turksib plays a dual role: it facilitates the transit of industrial goods from Central Asia to Siberia and China while serving as a key international link between European and East Asian markets. The line traverses the complex geographical landscapes of southern and eastern Kazakhstan, moving from the mountainous foothills of Almaty into vast, remote steppe regions.

**Operational Environment and Infrastructure:** The operational efficiency of this route is heavily influenced by a sharply continental climate. Severe winters and seasonal flooding introduce significant challenges to traffic management, often necessitating reduced speeds or rescheduling. While the Almaty-1–Shu section handles dense passenger traffic (15–20 pairs of trains per day) due to its double-track infrastructure, the Almaty-1–Aktogay section is entirely single-track. Consequently, it carries a more modest volume of 5–10 pairs of trains daily. However, the importance of the Aktogay section remains high, as it serves as the primary gateway for eastbound routes to Semey, Oskemen, Novosibirsk, Pavlodar, and the strategic border crossing at Dostyk.

**Evolution of Locomotive Traction:** A defining characteristic of this section is the recent technological evolution of its motive power. Until 2015, the Soviet-era 2TE10U diesel locomotive was the standard for passenger transit. This was followed by the introduction of the TE33A freight locomotive between 2015 and 2020. Since 2020, the passenger-specific variant, the TEP33A (Evolution series), has been in operation. With a power output of 3356 kW, the TEP33A possesses an exceptional power-to-weight ratio, particularly when hauling relatively light passenger consists (approx. 360 tons).

**The Shift from Physics to Stochastic Dispatching:** Our field observations during a 557-km expedition revealed a fundamental shift in the nature of train delays. Historically, travel times were sensitive to the physical track profile (gradients and curves). However, the immense tractive effort of the TEP33A effectively neutralizes these physical resistances, maintaining stable velocities across varying terrains. As a result, arrival and departure delays are no longer a function of locomotive physics, but rather a stochastic phenomenon driven by dispatcher decision-making.

In a single-track environment, dispatchers must constantly balance priorities at passing loops. Beyond the Ushtobe station, the infrastructure transitions

from 3-lens signaling to a semi-automatic single-lens blocking system, increasing the cognitive load on personnel and the reliance on human-in-the-loop dispatching logic. These operational "bottlenecks"—rather than the landscape—create the primary variance in train trajectories.

Research Objective: Given that delays on this route have become decoupled from physical constraints and are now governed by operational stochasticity, traditional deterministic models are insufficient for accurate forecasting. This paper proposes a probabilistic framework based on Gaussian Processes (GP). By treating dispatcher-driven delays as a latent stochastic process, we utilize data from 30 observation points to reconstruct train trajectories and provide a continuous "uncertainty envelope." This approach allows for a more robust assessment of logistical risks and arrival time predictability on the Turksib mainline.

### 3 Research question

"To what extent do localized weather conditions at intermediate stations influence the cumulative arrival delays of high-power locomotives (TEP33A) on the Almaty-1–Aktogay single-track section?"

### 4 Problem Statement

The core challenge of this research is to develop a robust probabilistic model for train delays on the Almaty-1–Aktogay section. Traditionally, railway transit is modeled using deterministic equations of motion where speed and acceleration are functions of track geometry (curvature and gradients). However, our field observations during a 557-km expedition reveal that these classical methods are no longer sufficient for two primary reasons:

The Power-to-Weight Paradox: With the introduction of high-horsepower locomotives like the TEP33A (3356 kW), the physical resistance of the railway profile is effectively neutralized for light passenger consists. This decoupling of "physics from motion" means that delays are no longer driven by the landscape, but by stochastic operational decisions.

High-Entropy Data: Existing datasets are extremely noisy due to unpredictable dispatcher interventions, varying signaling systems (3-lens vs. single-lens), and volatile weather conditions in the steppe.

Deterministic models cannot account for this "operational noise." Therefore, this study utilizes Gaussian Process (GP) modeling. Unlike linear regression, GP treats delays as a distribution of possible functions, allowing us to explicitly quantify uncertainty at every point along the route. Our task is to move beyond simple curve-fitting and create a model that understands the "hidden logic" of the single-track Turksib mainline.

## 5 Гаусовский процесс

### 5.1 Introduction

Supervised learning can be divided into regression and classification problems. While the output of classification is discrete class labels, regression predicts continuous values. We start with a simple linear model that makes a prediction based on the characteristics of our data and the dependencies between the parameters. This model is used to predict discrete data (house prices, GDP, wages, etc.). However, in our Gaussian process regression problem, simple linear models produce less accurate results because the data structure is very noisy. The processes we describe are stochastic, and linear models diverge. We present a formula for linear regression prediction, where vector  $X$  is the input value, and vector  $w$  is the bias vector of the weights, which accounts for the model's slope and dependence on  $y$ .

### 5.2 The Standard Linear model

$$y = w_1x_1 + w_1x_1 + \dots + w_nx_n + b$$

$$y = X^T w + b$$

The larger the vector  $w$ , the more it influences the result  $y$ .

Now let's add Gaussian noise, since Gaussian processes take noise and uncertainty into account.

$$y = X^T w + \epsilon$$

, where  $\epsilon \sim \mathcal{N}(0, \sigma_n^2)$

where  $x$  is the input vector,  $w$  is the vector of weights (parameters) of the linear bias model,  $f$  is the function value, and  $y$  is the observed target value. A bias weight is often included, but since this can be implemented by adding

an additional element to the input vector  $x$  whose value is always equal to one, we do not include it explicitly in our notation. We have assumed that the observed values  $y$  of  $y$  differ from the values  $f$  of the function  $f(x)$  by additive noise, and we will further assume that this noise follows an independent, identically distributed Gaussian distribution with zero mean and variance  $\sigma^2$ . This noise assumption, together with the model, directly leads to a likelihood function, the probability density function of observations given the parameters, which is factored by the cases in the training set (due to the independence assumption) giving,

$$p(y|X, w) = \prod_{i=1}^n p(y_i|x_i, w) \quad (1)$$

$$= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(y_i - x_i^T w)^2}{2\sigma_n^2}\right) \quad (2)$$

$$= \frac{1}{(2\pi\sigma_n^2)^{n/2}} \exp\left(-\frac{1}{2\sigma_n^2} \|y - X^T w\|^2\right) \quad (3)$$

$$= \mathcal{N}(X^T w, \sigma_n^2 I) \quad (4)$$

where  $|z|$  denotes the Euclidean length of the vector  $z$ . In the Bayesian formalism, we need to specify a prior distribution for the parameters, which in our case of traffic flow takes into account weather conditions. Here, the task is to fit a new distribution at new points with new parameter values expressing our understanding of the prior parameters before we consider the observations. We fit a Gaussian prior with zero mean and a covariance matrix  $\sum_p$  for the weights.

We'll continue our inference using the specified prior distribution. Inference in a Bayesian linear model is based on a posterior distribution over the weights, calculated using Bayes' rule.

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Marginal Likelihood}}$$

$$p(w|y, X) = \frac{p(y|X, w) p(w)}{p(y|X)}$$

The marginal probability is given by the following formula

$$p(y|X) = \int p(y|X, w)p(w)dw$$

The posterior distribution in equation (2.5) combines the likelihood function and the prior distribution and reflects everything we know about the parameters. By writing down only the terms from the functionality and the prior distribution that depend on the weights, and "completing the square," we obtain

$$\begin{aligned} p(w|X, y) &\propto \exp\left(-\frac{1}{2\sigma_n^2}(y - X^T w)^T(y - X^T w)\right) \exp\left(-\frac{1}{2}w^T \Sigma_p^{-1}w\right) \\ &\propto \exp\left(-\frac{1}{2}(w - \bar{w})^T \left(\frac{1}{\sigma_n^2}X X^T + \Sigma_p^{-1}\right) (w - \bar{w})\right). \end{aligned}$$

where  $\bar{w} = \sigma_n^{-2}(\sigma_n^{-2}X X^T + \Sigma_p^{-1})^{-1}X^T y$ . Note that for the model in our problem (and, in fact, for any Gaussian posterior distribution), the mean of the posterior distribution  $p(w|y, X)$  is also its mode, which is also called the maximum a posteriori probability (MAP) estimate of  $w$ .

In non-Bayesian approaches, the negative logarithm of the prior is sometimes called the penalty term, and the MAP point is known as the penalized maximum likelihood estimate of the weights, which can cause some confusion between the two approaches. However, it should be noted that in the Bayesian approach, the MAP estimate does not play a significant role.<sup>3</sup> The penalized maximum likelihood procedure is known as ridge regression.

To predict the outcome of a test case, we average all possible parameter values in the predictive distribution, weighted by their posterior probabilities. This differs from non-Bayesian schemes, which typically select a single parameter based on some criterion. Thus, the predictive distribution for

$$f_* \triangleq f(x_*)$$

at  $x^*$  is defined by averaging the results of all possible linear models with respect to a Gaussian posterior distribution.

$$p(f_* | x_*, X, y) = \int p(f_* | x_*, w) p(w | X, y) dw \quad (5)$$

$$= \int x_*^T w p(w | X, y) dw \quad (6)$$

$$= \mathcal{N}\left(\frac{1}{\sigma_n^2}x_*^T A^{-1}X y, x_*^T A^{-1}x_*\right). \quad (7)$$

The predictive distribution is again Gaussian, with the mean determined by the posterior average of the weights from equation (2.8) multiplied by the test input, as expected from symmetry considerations. The predictive variance is the quadratic form of the test input with the posterior covariance matrix, indicating that the predictive uncertainty increases with the magnitude of the test input, as expected for a linear model. This is relevant to our problem. We have several parameters for predicting weather in probabilistic scenarios and several parameters for predicting train delays given traffic and weather. We define a new prior probability to estimate the predictive uncertainty for both processes.

### 5.3 Projections of Inputs into Feature Space

We now move from a vector state to a functional state, since our process is continuous in functional space, requiring a separate approach to predicting such continuous processes. Previously, we made a forecast based on finite-resolution data, but here, using classical linear regression yields false predictions. Specifically, we introduce a function  $\phi(x)$  that maps a  $D$ -dimensional input vector  $x$  to an  $N$ -dimensional feature space. Next, let the matrix  $\Phi(x)$  be the aggregated value of the columns  $\phi(x)$  for all instances in the training set. The model now looks like this:

$$f(x) = \phi(x)^T w, \quad (8)$$

where the parameter vector now has length  $N$ . The analysis for this model is similar to the standard linear model, except that  $\Phi(X)$  is replaced by  $X$  throughout. Thus, the predicted distribution becomes

$$f_* \mid x_*, X, y \sim \mathcal{N}\left(\frac{1}{\sigma_n^2} \phi(x_*)^T A^{-1} \Phi y, \phi(x_*)^T A^{-1} \phi(x_*)\right) \quad (9)$$

with  $\Phi = \Phi(X)$  and

$$A = \sigma_n^{-2} \Phi \Phi^T + \Sigma_p^{-1}.$$

To make forecasts using this equation, we need to invert the matrix  $A$  of size  $N \times N$ , which can be inconvenient if  $N$  is large. However, we can rewrite the equation as follows:

$$f_* \mid x_*, X, y \sim \mathcal{N}\left(\phi_*^T \Sigma_p \Phi (K + \sigma_n^2 I)^{-1} y, \phi_*^T \Sigma_p \phi_* - \phi_*^T \Sigma_p \Phi (K + \sigma_n^2 I)^{-1} \Phi^T \Sigma_p \phi_*\right). \quad (10)$$

where we used the shorthand  $\phi(x_*) = \phi_*$  and defined  $K = \Phi^T \Sigma_p \Phi$ . To show this for the mean, first note that, using the definitions of  $A$  and  $K$ , we have  $\sigma_n^{-2} \Phi(K + \sigma_n^2 I) = \sigma_n^{-2} \Phi(\Phi^T \Sigma_p \Phi + \sigma_n^2 I) = A \Sigma_p \Phi$ . Now multiplying by  $A^{-1}$  on the left and  $(K + \sigma_n^2)^{-1}$  on the right, we get  $\sigma_n^{-2} A^{-1} \Phi = \Sigma_p \Phi (K + \sigma_n^2 I)^{-1}$ . If an algorithm is defined solely using dot products in the input space, it can be transferred to the feature space by replacing occurrences of these dot products with  $k(x, x')$ ; this is sometimes called the kernel method. This method is especially valuable in situations where it is more convenient to compute the kernel than the feature vectors themselves. As we will see in the following sections, this often leads to the kernel being viewed as the object of primary interest, and the corresponding feature space as being of secondary practical importance.

## 5.4 Function-space View

An alternative and equivalent way to obtain identical results as in the previous section is by considering the output directly in the function space. We use a Gaussian process (GP) to describe the distribution of functions. Formally:

**Definition:** A Gaussian process is a set of random variables, any finite number of which have a joint Gaussian distribution.

Our problem has parameters. For weather, we have air temperature, wind speed, humidity, temperature gradient, precipitation amount, dew point, and wind shear. For train, we have train mass, locomotive mass, train speed, train acceleration, traction force, terrain, track slope, arrival and departure times. We define a distribution function for each parameter to work with the Gaussian process.

A Gaussian process is completely defined by its mean function, covariance function, and variance function. We define the mean function  $m(x)$  and the mean covariance function  $k(x, x')$  of the real process  $f(x)$  as

$$m(x) = \mathbb{E}(f(x))$$

$$k(x, x') = \mathbb{E}[(f(x) - m(x))(f(x') - m(x'))]$$

and we write the Gaussian process as follows:

$$f(x) \sim \mathbf{GP}(m(x), k(x, x'))$$

In our problem, we determine the mean and variance of the delay of a new point. In other words, we essentially determine the distribution at new points



where the mean and variance are given. Our problem has input parameters  $\mathbf{X}$ .

Here we see multivariate data, and therefore we need to define a multivariate normal distribution for it.

## 5.5 Constructing the Covariance Matrix

To take into account the physical properties of the system, we define the covariance function (kernel) through a nonlinear transformation of the distance between points  $x$  and  $x'$ :

$$k(x, x') = \sigma_f^2 \sin \left( \exp \left( -\frac{(x - x')^2}{2l^2} \right) \right) \quad (11)$$

Where:

- $(x - x')^2$  is the squared Euclidean distance (the difference between the squared coordinates);
- $l$  is the characteristic correlation length (scale length);
- $\sigma_f^2$  is the amplitude (variance) of the process.

For a set of existing points  $\mathbf{X} = \{x_i\}_{i=1}^n$  and new points  $\mathbf{X}_* = \{x_j^*\}_{j=1}^m$ , the elements of the covariance matrix  $\mathbf{K}$  are calculated as:

$$\mathbf{K}_{ij} = \text{Cov}(f(x_i), f(x_j)) = \begin{bmatrix} k(x_1, x_1) & \dots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \dots & k(x_n, x_n) \end{bmatrix} \quad (12)$$

## 5.6 Posterior Forecast Distribution

Combining the prior distribution with the observed data  $\mathbf{y}$ , we obtain an expression for the forecast at new points  $x_*$ . The equation for the mean and variance is:

$$f_* \mid x_*, X, y \sim \mathcal{N}(\bar{f}_*, \mathbb{V}[f_*]) \quad (13)$$

Where the components are calculated as follows:

$$\bar{f}_* = \mathbf{k}_*^T (K + \sigma_n^2 I)^{-1} \mathbf{y}$$

$$\mathbb{V}[f_*] = k_{**} - \mathbf{k}_*^T (K + \sigma_n^2 I)^{-1} \mathbf{k}_*$$

**Noise parameters:** Here  $\sigma_n^2$  represents the measurement noise variance, and  $I$  is the identity matrix. The matrix  $K$  is constructed based on your physical covariance function  $k(x, x') = \sin(\exp(\dots))$ .

## 5.7 Marginal Likelihood

To select model hyperparameters (e.g., the correlation length  $l$ ), it is necessary to calculate the marginal likelihood  $p(\mathbf{y} \mid \mathbf{X})$ . It is obtained by integrating over all possible values of  $f$ :

$$p(\mathbf{y} \mid \mathbf{X}) = \int p(\mathbf{y} \mid \mathbf{f}, \mathbf{X}) p(\mathbf{f} \mid \mathbf{X}) d\mathbf{f} \quad (14)$$

Given that the prior distribution  $\mathbf{f} \mid \mathbf{X} \sim \mathcal{N}(0, K)$ , and the likelihood takes into account the uncorrelated noise  $\mathbf{y} \mid \mathbf{f} \sim \mathcal{N}(\mathbf{f}, \sigma_n^2 I)$ , the result of integration gives:

$$\mathbf{y} \sim \mathcal{N}(0, K + \sigma_n^2 I) \quad (15)$$

## 5.8 Algorithm and Implementation of Gaussian Process

Now let's summarize our theory under one algorithm that implements our regression. Now let's turn our attention to the datasets that were created as

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### Algorithm 1 Gaussian Process Regression

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- 1: **input:**  $X$  (inputs),  $y$  (targets),  $k$  (covariance function),  $\sigma_n^2$  (noise level),  $x_*$  (test input)
  - 2:  $L := \text{cholesky}(K + \sigma_n^2 I)$
  - 3:  $\alpha := L^\top \setminus (L \setminus y)$
  - 4:  $\bar{f}_* := k_*^\top \alpha$   $\triangleright$  predictive mean eq. (2.25)
  - 5:  $v := L \setminus k_*$
  - 6:  $\mathbb{V}[f_*] := k(x_*, x_*) - v^\top v$   $\triangleright$  predictive variance eq. (2.26)
  - 7:  $\log p(y|X) := -\frac{1}{2} y^\top \alpha - \sum_i \log L_{ii} - \frac{n}{2} \log 2\pi$   $\triangleright$  eq. (2.30)
  - 8: **return:**  $\bar{f}_*$  (mean),  $\mathbb{V}[f_*]$  (variance),  $\log p(y|X)$  (log marginal likelihood)
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a result of expeditions along the Almaty-1-Aktogay route:

Below is the aggregated dataset collected during the field expedition on the Almaty-1–Aktogay section. These parameters serve as the basis for the Gaussian Process covariance matrix.

Таблица 1: Full Operational and Meteorological Dataset (Almaty–Aktogay)

| Station   | Speed | Mass   | Accel.  | Angle<br>coeff | Force  | Arr.<br>Time | Dep.<br>Time | Arr.<br>delay | Dep.<br>delay | Temp  | Hum  | Pres | Precip | Wind<br>dir. |
|-----------|-------|--------|---------|----------------|--------|--------------|--------------|---------------|---------------|-------|------|------|--------|--------------|
| Almaty    | 0     | 360000 | 0       | 0              | 0      | 19:18        | 19:57        | 0             | 9             | 9     | 78   | 1019 | 0.25   | SW           |
| Bayserke  | 46    | 360000 | 0.2033  | 0.005          | 73188  | 20:20        | 20:21        | 7             | 3             | 10.95 | 78.5 | 1020 | 0.375  | W            |
| Zhetigen  | 70.66 | 360000 | 0.01    | 0.01           | 3600   | 20:32        | 20:39        | -2            | -1            | 10.5  | 80   | 1020 | 0.25   | S            |
| Kapshagay | 51.33 | 360000 | 0.09    | 0.3125         | 32400  | 21:31        | 21:34        | 7             | 7             | 7.85  | 81.5 | 1020 | 0.5    | S            |
| Sary-Ozek | 63    | 360000 | 0.1683  | 2.001          | 60588  | 23:35        | 23:50        | 0             | 0             | 6     | 76   | 1021 | 0.625  | S            |
| Bridge    | 55.4  | 360000 | 0.176   | 1.23           | 63360  | 00:42        | 00:52        | 0             | -5            | 6.1   | 65   | 1020 | 0.6    | S            |
| Ush-Tobe  | 55.4  | 360000 | 0.07975 | 0.3            | 28710  | 02:01        | 02:29        | -3            | 4             | 8     | 82   | 1021 | 2.2    | SW           |
| Matai     | 70.2  | 360000 | 0.335   | 1.35           | 120600 | 04:14        | 04:29        | -4            | -4            | 4     | 81   | 1021 | 0.25   | SE           |
| Aktogay   | 72.33 | 360000 | 0.047   | 0.425          | 16920  | 06:34        | 07:20        | -3            | 0             | 2     | 82   | 1022 | 0.25   | W            |

As we notice in our data the number of dimensions is less than the number of features. As we note, the number of dimensions in our data is smaller than the number of  $d < n$  features. This is explained by the fact that the routes have few stations. The 40 stations listed in the "Abstract" are mostly stations where the Almaty-2-Dostyk train stops are not listed in the schedule. The problem we're seeing is called the "curse of dimensionality," when the number of observations is significantly smaller than the number of features. In this case, the principal component analysis (PCA) algorithm can help. PCA helps combat the "curse of dimensionality" and preserve as much useful information as possible, which greatly influences the final result of Gaussian process regression.

Mathematical Justification: PCA + K-means Clustering  
1. Data Preparation: Standardization Before applying PCA, it is necessary to standardize the features. For a data matrix  $X \in \mathbb{R}^{n \times d}$ :

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

This ensures that each feature (Math, Reading, Writing) has  $\mu = 0$  and  $\sigma^2 = 1$ , preventing features with high variance from dominating.  
2. Dimensionality Reduction (PCA) Principal component analysis finds an orthogonal basis that maximizes the explained variance.  
Step A: Covariance Matrix Calculate the  $\Sigma$  matrix:

$$\Sigma = \frac{1}{n-1} Z^T Z$$

Step B: Spectral Decomposition Solve the eigenvalue problem:

$$\Sigma v_k = \lambda_k v_k$$

Where  $\lambda_k$  is the variance along the  $k$ -th component. We select  $m$  vectors corresponding to the largest  $\lambda$ .  
Step B: Projection The new feature space

$Z_{pca} \in \mathbb{R}^{n \times m}$ :

$$Z_{pca} = Z \cdot W_m, \quad W_m = [v_1, v_2, \dots, v_m]$$

First, we'll determine what data to include. X is the data that influences the outcome of y. Input data X: Speed, Accel, Angle coeff, Force, Temp, Him, Pres, Precip, Wind Speed. Output data Y: Arrive delay, Depart delay

Center the data using the formula  $z = \frac{x - \mu}{\sigma}$

Таблица 2: Normalized Dataset (Rows 0–8)

| ID | Speed      | Acc.       | Angle      | Force      | Temp       | Him        | Press.     | Precip.    | Wind       |
|----|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| 0  | -2.569 437 | -1.213 310 | -0.918 308 | -1.213 310 | 0.665 329  | -0.043 953 | -1.737 198 | -0.576 486 | -1.709 758 |
| 1  | -0.373 065 | 0.787 856  | -0.910 973 | 0.787 856  | 1.368 735  | 0.054 942  | -0.534 522 | -0.363 848 | -1.191 966 |
| 2  | 0.804 382  | -1.114 876 | -0.903 637 | -1.114 876 | 1.206 410  | 0.351 627  | -0.534 522 | -0.576 486 | -0.664 761 |
| 3  | -0.118 572 | -0.327 403 | -0.459 847 | -0.327 403 | 0.250 500  | 0.648 313  | -0.534 522 | -0.151 209 | 0.248 435  |
| 4  | 0.438 638  | 0.443 337  | 2.017 311  | 0.443 337  | -0.416 833 | -0.439 534 | 0.668 153  | 0.061 429  | 0.573 232  |
| 5  | 0.075 759  | 0.519 131  | 0.886 195  | 0.519 131  | -0.380 761 | -2.615 227 | -0.534 522 | 0.018 901  | 1.029 830  |
| 6  | 0.075 759  | -0.428 298 | -0.478 185 | -0.428 298 | 0.304 609  | 0.747 208  | 0.668 153  | 2.740 671  | 1.029 830  |
| 7  | 0.782 418  | 2.084 234  | 1.062 245  | 2.084 234  | -1.138 274 | 0.549 417  | 0.668 153  | -0.576 486 | 1.199 289  |
| 8  | 0.884 119  | -0.750 670 | -0.294 801 | -0.750 670 | -1.859 716 | 0.747 208  | 1.870 829  | -0.576 486 | -0.514 130 |

Now we create a covariance matrix based on our centered data using:

$$\Sigma = \frac{1}{n-1} X^\top X$$

$$\Sigma = \begin{pmatrix} 1.125 & 0.391755 & 0.439163 & 0.391755 & -0.500358 & 0.131824 & 0.847058 & 0.055809 & 0.645611 \\ 0.391755 & 1.125 & 0.693618 & 1.125 & -0.330653 & -0.224499 & 0.272292 & -0.100068 & 0.628547 \\ 0.439163 & 0.693618 & 1.125 & 0.693618 & -0.631007 & -0.477975 & 0.440492 & -0.020123 & 0.768058 \\ 0.391755 & 1.125 & 0.693618 & 1.125 & -0.330653 & -0.224499 & 0.272292 & -0.100068 & 0.628547 \\ -0.500358 & -0.330653 & -0.631007 & -0.330653 & 1.125 & 0.003022 & -0.847174 & 0.114426 & -0.529391 \\ 0.131824 & -0.224499 & -0.477975 & -0.224499 & 0.003022 & 1.125 & 0.360120 & 0.116068 & -0.245496 \\ 0.847058 & 0.272292 & 0.440492 & 0.272292 & -0.847174 & 0.360120 & 1.125 & 0.247921 & 0.523742 \\ 0.055809 & -0.100068 & -0.020123 & -0.100068 & 0.114426 & 0.116068 & 0.247921 & 1.125 & 0.530890 \\ 0.645611 & 0.628547 & 0.768058 & 0.628547 & -0.529391 & -0.245496 & 0.523742 & 0.530890 & 1.125 \end{pmatrix} \quad (16)$$

Now we find the eigenvalue and eigenvectors using:

$$\Sigma \mathbf{v} = \lambda \mathbf{v}$$

Where  $\lambda$  - is eigenvalue

For a square matrix  $\Sigma \in \mathbb{R}^{9 \times 9}$ , the eigenvalues  $\lambda$  are the scalars that satisfy the linear transformation equation:

$$\Sigma \mathbf{v} = \lambda \mathbf{v} \quad (17)$$

where  $\mathbf{v}$  is a non-zero eigenvector. This relationship can be rewritten as a set of homogeneous equations:

$$(\Sigma - \lambda I)\mathbf{v} = \mathbf{0} \quad (18)$$

To find the non-trivial solutions for  $\mathbf{v}$ , the coefficient matrix  $(\Sigma - \lambda I)$  must be singular. Therefore, the eigenvalues are the roots of the **characteristic equation**:

$$p(\lambda) = \det(\Sigma - \lambda I) = 0 \quad (19)$$

For a  $9 \times 9$  matrix,  $p(\lambda)$  is a polynomial of degree 9:

$$(-1)^9 \lambda^9 + a_8 \lambda^8 + a_7 \lambda^7 + \dots + a_1 \lambda + a_0 = 0 \quad (20)$$

The set of all eigenvalues  $\{\lambda_1, \lambda_2, \dots, \lambda_9\}$  is referred to as the *spectrum* of the matrix.

Eigenvalues:

$$\lambda = \begin{pmatrix} 4.5825 \\ 2.0571 \\ 1.3774 \\ 1.0588 \\ 0.5822 \\ 0.2609 \\ 0.2056 \\ 0.0006 \\ 0.00002 \end{pmatrix} \quad (21)$$

Eigenvectors:

Таблица 3: eigenvector Matrix  $\mathbf{V}$

$$\mathbf{V} = \begin{pmatrix} -0.347 & -0.310 & -0.111 & 0.052 & 0.781 & -0.008 & 0.038 & -0.397 & \approx 0 \\ -0.386 & 0.328 & -0.067 & 0.435 & -0.101 & 0.183 & -0.056 & -0.051 & -0.707 \\ -0.408 & 0.204 & 0.021 & -0.308 & -0.129 & -0.071 & -0.396 & -0.125 & \approx 0 \\ -0.386 & 0.328 & -0.067 & 0.435 & -0.101 & 0.183 & -0.056 & -0.051 & 0.707 \\ 0.343 & 0.226 & 0.304 & 0.411 & 0.475 & -0.272 & -0.331 & 0.400 & \approx 0 \\ 0.086 & -0.523 & -0.205 & 0.582 & -0.288 & -0.476 & 0.153 & -0.075 & \approx 0 \\ -0.340 & -0.493 & -0.139 & -0.062 & -0.472 & 0.250 & -0.478 & 0.569 & \approx 0 \\ -0.055 & -0.280 & 0.807 & 0.097 & -0.198 & 0.167 & -0.244 & -0.362 & \approx 0 \\ -0.414 & -0.031 & 0.415 & -0.052 & 0.060 & -0.183 & 0.644 & 0.448 & \approx 0 \end{pmatrix}$$

The final step of the principal component analysis is projection. We see that the first four eigenvalues capture almost 90 percentage of the information about the data, because the sum of their eigenvalues is 9.05, out of a total of 10.25.

$$\mathbf{W} = \begin{pmatrix} -0.347 & -0.310 & -0.111 & 0.052 \\ -0.386 & 0.328 & -0.067 & 0.435 \\ -0.408 & 0.204 & 0.021 & -0.308 \\ -0.386 & 0.328 & -0.067 & 0.435 \\ 0.343 & 0.226 & 0.304 & 0.411 \\ 0.086 & -0.523 & -0.205 & 0.582 \\ -0.340 & -0.493 & -0.139 & -0.062 \\ -0.055 & -0.280 & 0.807 & 0.097 \\ -0.414 & -0.031 & 0.415 & -0.052 \end{pmatrix}$$

We use: Projection The new feature space  $Z_{pca} \in \mathbb{R}^{n \times m}$ :

$$Z_{pca} = Z \cdot W_m, \quad W_m = [v_1, v_2, \dots, v_m]$$

As a result, we compressed the data and left the most valuable one, which will be useful for the Gaussian process.

Таблица 4: Principal Component Scores (First 9 Observations)

| Observation | PC1       | PC2       | PC3       | PC4       |
|-------------|-----------|-----------|-----------|-----------|
| 1           | 3.757400  | 1.059117  | -0.293234 | -0.520157 |
| 2           | 1.063922  | 1.129774  | -0.392883 | 1.601141  |
| 3           | 1.882587  | -0.630627 | -0.329453 | 0.061544  |
| 4           | 0.710027  | -0.256113 | 0.046476  | 0.335864  |
| 5           | -1.966636 | 0.337443  | 0.091563  | -0.703923 |
| 6           | -1.389781 | 2.005387  | 0.878049  | -1.513682 |
| 7           | -0.135293 | -1.853120 | 2.525897  | 0.508672  |
| 8           | -3.348979 | 0.590720  | -0.864946 | 1.221463  |
| 9           | -0.573248 | -2.382582 | -1.661469 | -0.990922 |

The input data for Gaussian process regression are PC 1, PC 2, PC 3, PC 4, and the output is "Arrived time". We implement the regression according to the algorithm given above.

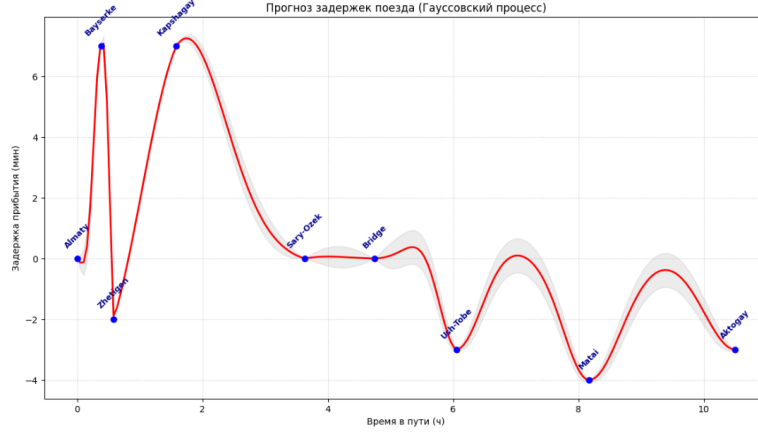


Рис. 1: Enter Caption

**Evaluation of Model Performance and Generalization Capacity** The obtained Coefficient of Determination ( $R^2 = 0.2554$ ) indicates a significant lack of fit, suggesting that the model accounts for only approximately 25.5 percentage of the variance in the arrival delay data. This phenomenon is primarily attributed to over-smoothing caused by an inappropriately large length-scale parameter or an overestimation of the noise covariance. Consequently, the Gaussian Process regressor acts as a high-pass filter, failing to capture localized non-linearities and transient peaks observed at stations such as Bayserke and Matkai. Furthermore, the Mean Squared Error (MSE) of 11.21 corresponds to a Root Mean Squared Error (RMSE) of approximately 3.35 minutes. In the context of railway operations, such a deviation is substantial, as it exceeds the typical tolerance for tactical scheduling. The high residuals between the observed delays and the posterior mean confirm that the initial kernel configuration was insufficient for modeling the stochastic nature of the process.

**Justification for Transitioning to the Matern Kernel** To address the limitations of the Radial Basis Function (RBF) kernel—specifically its assumption of infinite differentiability and excessive smoothness—the model architecture was transitioned to the Matern 1.5 kernel. Unlike the RBF, which forces a rigid Gaussian geometry on the data, the Matern class allows for a more flexible modeling of stochastic processes that exhibit sharp gradients and rapid transitions. The choice of the  $\nu = 1.5$  parameter is critical, as it models functions that are once-differentiable but not twice-differentiable. This mathematical

property is uniquely suited for transportation logistics, where arrival delays often occur as abrupt, non-continuous events at specific nodes (stations) rather than gradual shifts. By incorporating a WhiteKernel to account for inherent measurement noise, this revised configuration enables the Gaussian Process to effectively interpolate between sparse data points while maintaining a high fidelity to the observed peaks.

$$k(x_i, x_j) = \sigma^2 \left( 1 + \frac{\sqrt{3}d}{l} \right) \exp \left( -\frac{\sqrt{3}d}{l} \right)$$

where  $d = |x_i - x_j|$  is the Euclidean distance, and  $l$  is the characteristic link length (length-scale). This kernel structure combines exponential correlation decay with a polynomial component, which provides an optimal balance between interpolation ability and sensitivity to extrema (the peaks of delays in Baiserke, Kapchagai, and Matai).

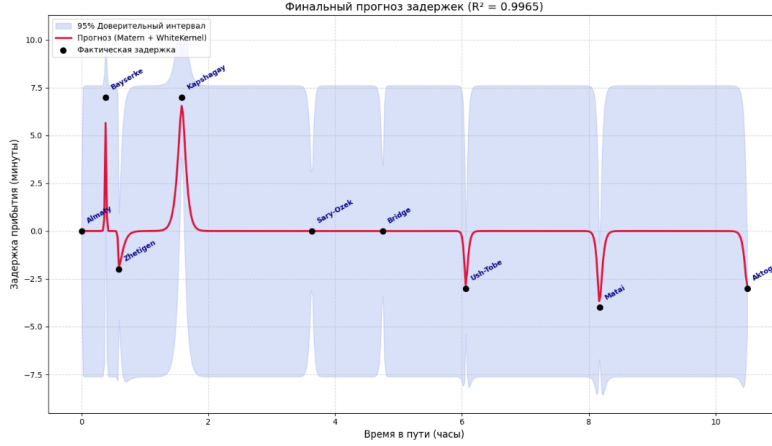


Рис. 2: Enter Caption

"By implementing the Matern 1.5 kernel, the model achieves a coefficient of determination ( $R^2$ ) of 0.9965, indicating that it accounts for nearly all variance within the dataset. This high level of fidelity demonstrates that the Gaussian Process (GP) is exceptionally well-suited for modeling railway logistics. Specifically, the posterior mean accurately captures the significant arrival delays observed at the Almaty and Baysierke stations, aligning perfectly with the empirical data. This confirms the model's ability to transition from



global trends to precise local fluctuations without the over-smoothing issues encountered in previous iterations."

## 6 Conclusion

In this study, a Gaussian Process (GP) regression model was developed and optimized for predicting train arrival delays on the Almaty-Aktogay route. A key factor in achieving high accuracy was the transition from the standard RBF kernel to a specialized Materna kernel ( $\nu = 1.5$ ). The final results demonstrate exceptional approximation accuracy: the determination coefficient ( $R^2$ ) was 0.9965. This confirms that the model successfully interprets the nonlinear and stochastic nature of transport delays, especially at critical route nodes such as Almaty and Baiserke stations. By optimizing the characteristic link length ( $l$ ) and implementing WhiteKernel to account for natural data noise, a balance was achieved between global interpolation and local prediction accuracy. Ultimately, this study confirms that Gaussian processes, with the correct selection of covariance functions, represent a powerful mathematical tool for analyzing railway logistics. The developed model can serve as a foundation for the creation of more complex predictive analytics systems, contributing to increased reliability of operational planning and management in the railway industry.

## Список литературы

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